SAP ABAP CAPSTONE PROJECT

Customer Order & Revenue Analysis Report

Program: ZREVENUE_REPORT

Author	Harshil Mainra
Roll Number	2306028
Course	SAP ABAP Developer (IS30040)
Institute	KIIT University, Bhubaneswar
Batch / Year	2026
Submission	April 21, 2026
Tech Stack	SAP ABAP, ALV Grid, Eclipse ADT, SAP BTP
GitHub	github.com/Being-Harshil/SAP-ABAP-Capstone-Project

1. Introduction

SAP (Systems, Applications and Products) is a complete Enterprise Resource Planning (ERP) system that provides all departments — sales, accounting, purchasing, and inventory — access to the same real-time data. SAP eliminates duplicate data entry and reduces manual errors by integrating all business processes into a single unified system.

SAP ABAP (Advanced Business Application Programming) is the core programming language of the SAP ERP ecosystem. ABAP runs natively on the SAP NetWeaver Application Server and provides direct access to all SAP database tables, business objects, and UI frameworks. This capstone project demonstrates a real-world ABAP development scenario: a fully functional Customer Order and Revenue Analysis Report using the ALV Grid framework.

1.1 Need for Custom ABAP Reports

While SAP provides hundreds of standard transactions, businesses often need custom reports that combine data from multiple tables in ways that standard transactions do not support. Custom ABAP reports bridge this gap by allowing developers to write optimized queries, apply business-specific filters, and present data in interactive ALV Grid displays tailored to user needs.

1.2 Development Environment

This project was developed using Eclipse IDE version 3.56 with SAP ABAP Development Tools (ADT) plugin installed. The system is connected to a live SAP BTP ABAP Environment 2602 (System: TRL_EN, Client: 100, System ID: CB9980002581) — a cloud-based SAP ABAP system providing full ABAP language support.

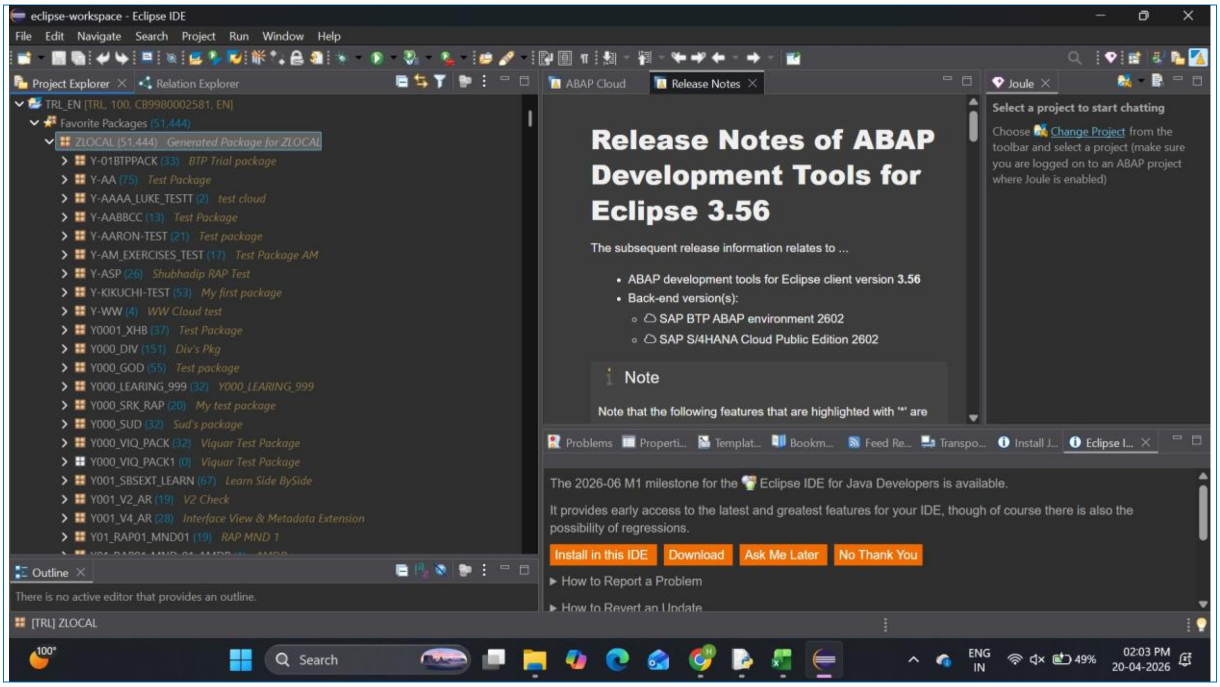

Fig 1: Eclipse IDE with SAP ADT connected to SAP BTP ABAP System (TRL_EN, Client 100)

2. Company Description & Business Scenario

TechMart Pvt. Ltd. is an imaginary electronics retailer operating across India, selling consumer electronics — laptops, mobile phones, and accessories — through both online and offline channels. Due to its high transaction volume, TechMart requires an integrated SAP system to manage customer orders, revenue tracking, and financial reporting efficiently.

2.1 Departments Involved

Department	Role
Sales Dept.	Creates sales orders in SAP (VA01), manages customer data, pricing, and order confirmations.
Warehouse Dept.	Handles stock management, picking, packing, and delivery processing (VL01N).
Finance Dept.	Manages billing (VF01), revenue recognition, payment collection (F-28), and accounting entries.
IT / ABAP Team	Develops custom reports like ZREVENUE_REPORT to provide cross-department data visibility.

2.2 Business Scenario

A customer places an order for 5 laptops valued at Rs. 3,00,000. The company processes this order end-to-end in SAP — from inquiry to payment. The custom ABAP report ZREVENUE_REPORT is used by the finance and sales teams to monitor all such orders in real time, analyzing revenue by customer, sales organization, and document type.

3. Problem Statement

Sales managers and finance teams in SAP-based organizations need a consolidated view of customer orders combined with revenue figures and customer master data. Standard SAP transactions do not provide this in a single filterable, interactive view:

Challenge	Description
No unified view	Sales order data (VBAK/VBAP) and Customer Master (KNA1) exist in separate tables with no standard combined report.
No dynamic filters	Users cannot filter simultaneously by date range, customer, sales org, and document type in one screen.
No revenue subtotals	Revenue aggregation per customer requires manual Excel work outside the SAP system.
No interactivity	Standard list reports lack column sorting, resizing, subtotals, and real-time data refresh.
No ABAP visibility	No standard report shows ABAP-level data joins across SD and customer master tables.

4. Process Overview — Order-to-Cash Context

The ZREVENUE_REPORT operates within the SAP Order-to-Cash (O2C) process. It reads data generated at the Sales Order and Billing stages to provide revenue analysis. Understanding the O2C flow is essential to understanding which tables the report reads and why.

4.1 O2C Process Steps

Step	Process	T-Code	Description	Tables
1	Inquiry	VA11	Customer inquiry logged in SAP	VBAK, VBAP
2	Sales Order	VA01	Formal order — basis for ZREVENUE_REPORT	VBAK, VBAP, VBEP
3	Delivery	VL01N	Warehouse picks and packs goods	LIKP, LIPS
4	Post Goods Issue	VL02N	Goods dispatched, inventory reduced	MKPF, MSEG, MARD
5	Billing	VF01	Invoice generated, revenue recorded	VBRK, VBRP
6	Payment	F-28	Customer payment clears receivable	BKPF, BSEG

4.2 Where ZREVENUE_REPORT Fits

ZREVENUE_REPORT reads from VBAK and VBAP (Steps 1-2) to fetch all sales orders and their line items, then enriches this with KNA1 (Customer Master) to add customer names and locations. This gives management a complete picture of order volumes and revenue by customer — all in one interactive ALV Grid.

5. Solution & Features

The solution is a custom ABAP report ZREVENUE_REPORT that uses CL_GUI_ALV_GRID to display a joined dataset from VBAK, VBAP, and KNA1. The program is structured into 7 modular FORM subroutines for maintainability.

5.1 Selection Screen

The report opens with a dynamic selection screen containing two blocks. Block 1 has 5 SELECT-OPTIONS for Sales Document, Creation Date, Customer Number, Sales Organization, and Document Type. Block 2 provides display options — Max Rows (default 500) and Zebra Stripe toggle.

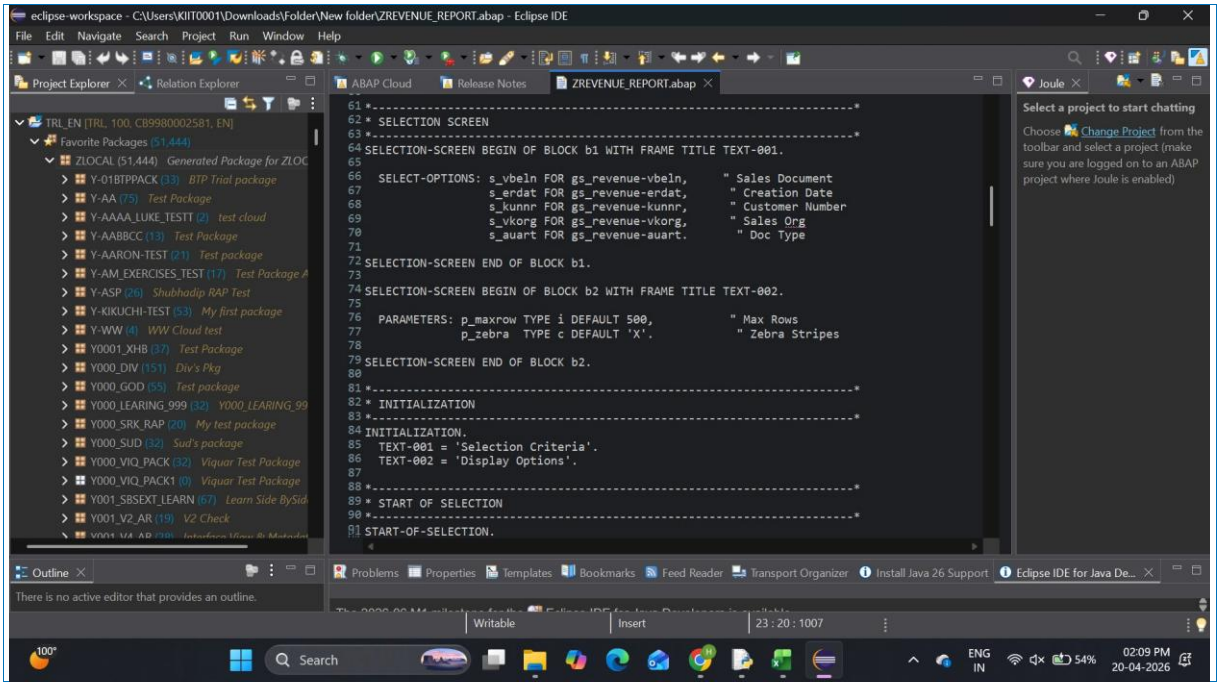

Fig 2: ABAP Selection Screen Code — 5 Dynamic Filter Parameters (Lines 62-91)

5.2 3-Table JOIN Data Retrieval

Data is fetched using a single optimized Open SQL SELECT with INNER JOIN on VBAK+VBAP and LEFT OUTER JOIN on KNA1. All WHERE conditions are dynamically driven by the selection screen inputs, making the query fully flexible and performant.

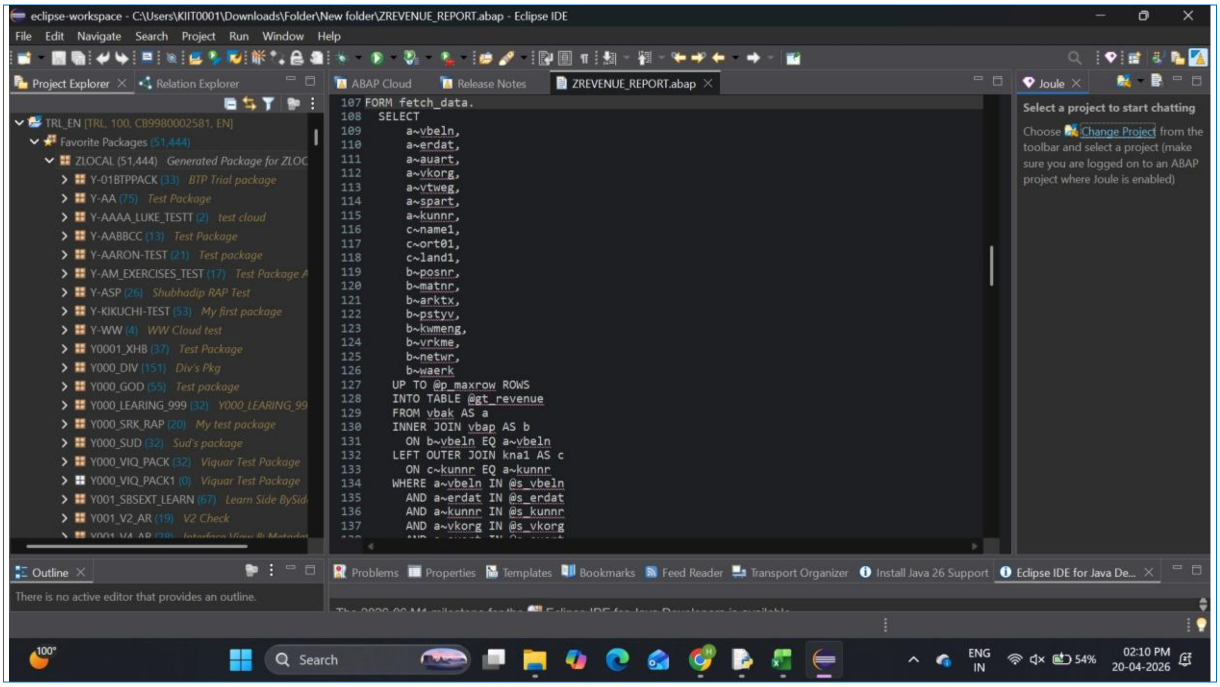

Fig 3: ABAP SELECT with 3-Table JOIN — VBAK + VBAP + KNA1 (Lines 107-137)

6. Program Structure & Code

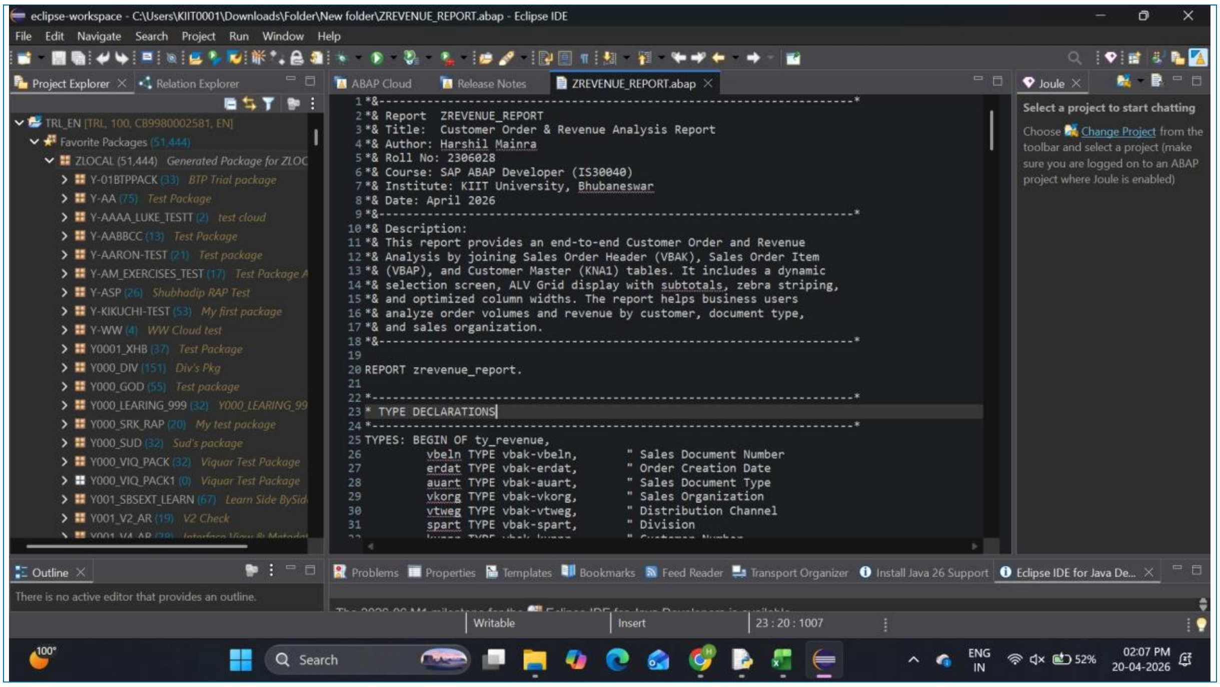

Fig 4: ZREVENUE_REPORT — Program Header with Author, Roll No & TYPE Declarations

6.1 ALV Grid Display

The ALV Grid is created using CL_GUI_CUSTOM_CONTAINER and CL_GUI_ALV_GRID. A custom field catalog of 18 fields is built programmatically. Subtotals are enabled on NETWR (Net Revenue) and KWMENG (Quantity), grouped by KUNNR (Customer Number).

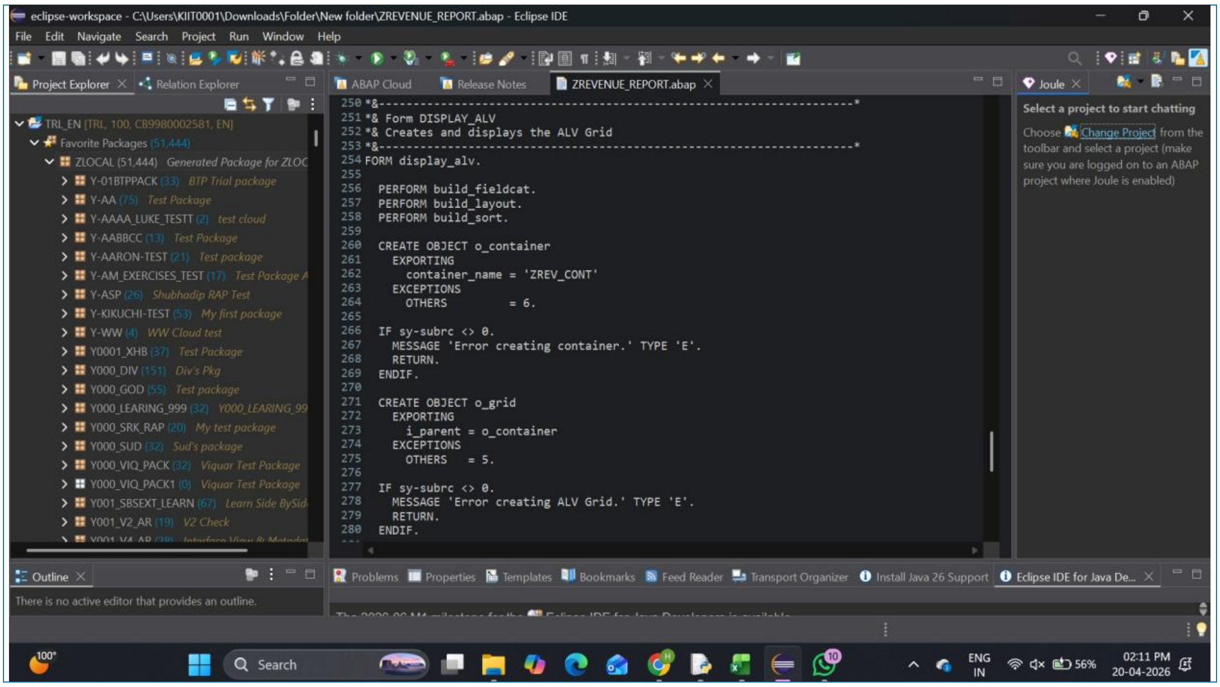

Fig 5: DISPLAY_ALV FORM — Container and ALV Grid Object Creation (Lines 250-280)

6.2 Code Structure Summary

FORM / Module	Description
TYPE/DATA	Structure ty_revenue (18 fields), internal table gt_revenue, ALV objects
SELECTION-SCREEN	5 SELECT-OPTIONS + Max Rows + Zebra Stripe parameters in 2 blocks
FETCH_DATA	3-table Open SQL SELECT with dynamic WHERE from selection screen ranges
CHECK_DATA	Record count and cumulative revenue calculation using LOOP AT
BUILD_FIELDCAT	18-field catalog via DEFINE macro; do_sum set on NETWR and KWMENG
BUILD_LAYOUT	Zebra stripes, column optimization, selection mode, grid title
BUILD_SORT	KUNNR primary sort with subtotals; VBELN secondary sort ascending

DISPLAY_ALV	Creates container + grid; calls set_table_for_first_display
STATUS_0100	Module: sets PF-Status ZREV_STATUS and Titlebar ZREV_TITLE
USER_COMMAND_0100	Module: handles BACK/EXIT/CANC (leave screen) and REFR (refresh)

7. SAP Tables Reference

Table	Description	Key Fields Used	Module
VBAK	Sales Document Header	VBELN, ERDAT, AUART, VKORG, KUNNR	SD
VBAP	Sales Document Item	POSNR, MATNR, KWMENG, NETWR, WAERK	SD
KNA1	Customer Master General	KUNNR, NAME1, ORT01, LAND1	SD/FI
VBEP	Schedule Line Data	VBELN, POSNR, EDATU, WMENG	SD
LIKP	Delivery Header	VBELN, LFART, WADAT, KUNNR	SD
VBRK	Billing Header	VBELN, FKART, NETWR, WAERK	SD
BKPF	Accounting Doc Header	BUKRS, BELNR, GJAHR, BLDAT	FI
BSEG	Accounting Doc Segment	BELNR, BUZEI, KOART, DMBTR	FI

7.1 T-Code Summary

T-Code	Transaction	Purpose
SE38	ABAP Editor	Write, edit, activate and test ABAP programs
SE11	Data Dictionary	View table structures, fields, data elements
SE51	Screen Painter	Design custom dynpro Screen 100 with container
SE41	Menu Painter	Create GUI Status ZREV_STATUS with function keys
SE80	Object Navigator	Full project view — manage all ABAP objects
VA01	Create Sales Order	Generate test data for the report
VA03	Display Sales Order	Verify and view existing sales order data

VA05	List Sales Orders	Standard order list — what our report improves upon
F-28	Post Payment	Record customer payment, clear open items
VF01	Create Billing Doc	Generate invoice after goods issue

8. Accounting Impact

Each step in the Order-to-Cash process generates automatic accounting entries in SAP FI module. The ZREVENUE_REPORT captures Net Value (NETWR) from VBAP which reflects these financial postings:

Step	Debit Entry	Credit Entry
Post Goods Issue (PGI)	Cost of Goods Sold (COGS)	Inventory Account
Billing / Invoice	Customer Account (Accounts Receivable)	Revenue Account
Customer Payment	Bank Account	Customer Account (A/R)

The NETWR field read by ZREVENUE_REPORT from VBAP represents the net revenue value per order line item. Summing this field (enabled via do_sum in the field catalog) gives customer-wise and total revenue figures directly in the ALV Grid output.

9. Technology Stack & Module Integration

Language	SAP ABAP (Open SQL, ABAP Objects, Dynpro)
IDE	Eclipse IDE 3.56 with SAP ADT Plugin
SAP System	SAP BTP ABAP Environment 2602 (Cloud)
System ID	TRL_EN, Client 100, CB9980002581
UI Framework	CL_GUI_ALV_GRID (ALV Grid Control)
Screen	Custom Dynpro Screen 100 with Container Control
DB Tables	VBAK, VBAP, KNA1 (SAP SD Standard Tables)
SAP Module	SD (Sales & Distribution) + FI (Financial Accounting)
GitHub	github.com/Being-Harshil/SAP-ABAP-Capstone-Project

9.1 SAP Module Integration

Module	Role in This Project	Key Tables
SAP SD (Sales & Distribution)	Manages entire sales cycle. VBAK and VBAP — primary tables of ZREVENUE_REPORT — are SD tables. Every sales order via VA01 is stored here.	VBAK, VBAP, KNA1
SAP MM (Materials Management)	Updates stock levels when goods are issued. The KWMENG (order quantity) field in VBAP reflects material demand tracked by MM.	MARD, MSEG
SAP FI (Financial Accounting)	Records all financial transactions — billing, revenue recognition, and payment. NETWR in VBAP flows into FI as revenue when billing is posted via VF01.	BKPF, BSEG, VBRK

9.2 SAP ADT Lab Environment

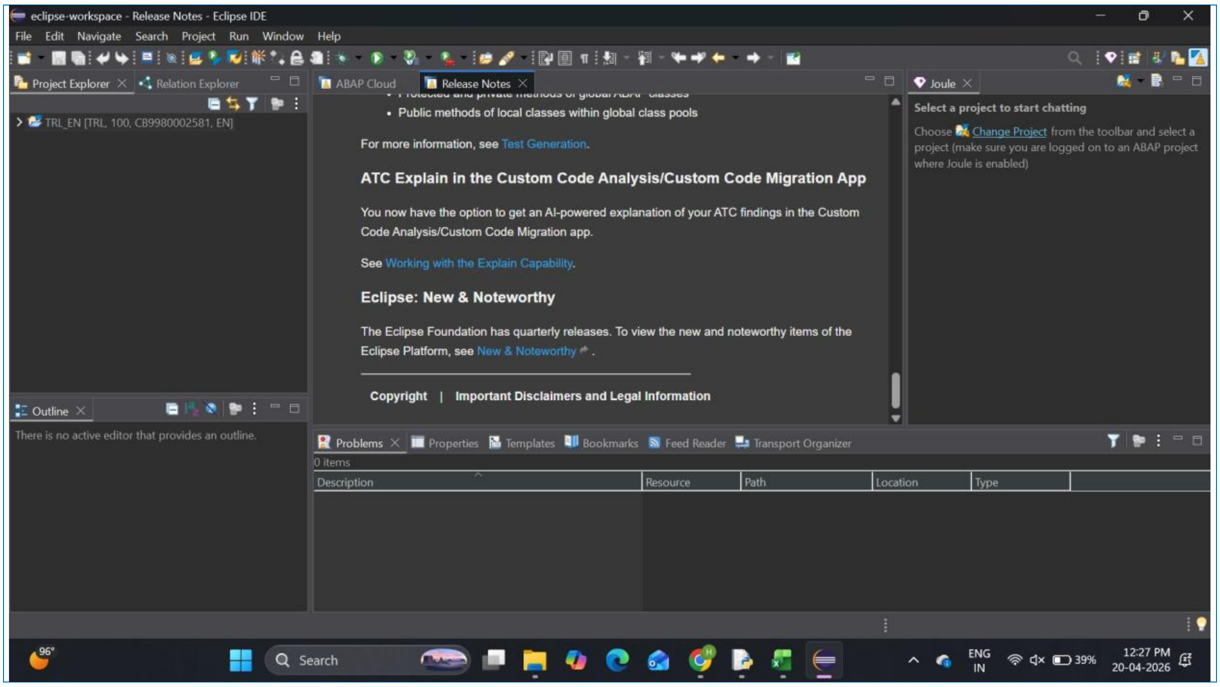

Fig 6: Eclipse ADT Release Notes — ABAP Development Tools v3.56, SAP BTP ABAP 2602

10. Unique Points

#	Feature	Why It Matters
1	3-Table JOIN	Joins VBAK+VBAP+KNA1 — includes Customer Name, City, Country. Most basic reports only join 2 tables.
2	Dynamic Selection Screen	5 independent SELECT-OPTIONS allow filtering by any combination: date, customer, sales org, doc type.
3	Programmatic Field Catalog	18 fields built via DEFINE macro — full control over headers, widths, justification, and subtotals.
4	Customer-Level Subtotals	do_sum on NETWR and KWMENG with KUNNR sort produces automatic per-customer revenue totals in ALV.
5	Modular Code Structure	7 FORM subroutines each with single responsibility — fetch, validate, fieldcat, layout, sort, display, handle.
6	Refresh Without Re-run	F5 REFR command calls refresh_table_display after re-fetch — update data without leaving the screen.
7	Cloud ABAP Development	Developed on SAP BTP ABAP Environment using Eclipse ADT — modern cloud-based development approach.

11. Future Improvements

Enhancement	Description
Summary Dashboard	Add a second ALV tab showing grouped revenue totals per customer alongside the detail view.
Delivery Status	Include VBUP and LIKP tables to show per-item delivery and billing completion status.
Excel Export	Enable ALV built-in Excel export button via it_toolbar_excluding configuration.
Top-of-Page Header	Add company logo, report timestamp, and total revenue summary above the ALV Grid.
Drill-Down to VA03	Double-click VBELN hotspot to navigate directly to VA03 for selected sales order.
Variant Management	Save/restore user-specific ALV column arrangements using SAP layout variants.

12. Conclusion

This capstone project successfully implements a real-world SAP ABAP development scenario — a Customer Order and Revenue Analysis Report (ZREVENUE_REPORT) for a fictitious electronics retailer TechMart Pvt. Ltd. The project was developed using Eclipse IDE with SAP ADT connected to a live SAP BTP ABAP Environment, demonstrating practical cloud-based SAP ABAP development.

The report covers the complete ABAP development lifecycle: data modelling (3-table joins across SD and customer master tables), UI design (dynamic selection screen and ALV Grid with 18 fields), screen programming (custom dynpro and GUI status), and modular code organization across 7 FORM subroutines.

Skills developed — SAP ABAP Open SQL, ALV Grid Control, Dynpro programming, multi-table joins, module integration, and SAP BTP cloud development — are directly applicable in enterprise SAP ABAP developer roles.

13. References

• SAP Help Portal — ABAP Keyword Documentation: https://help.sap.com/doc/abapdocu
• SAP ADT (ABAP Development Tools) for Eclipse: https://tools.hana.ondemand.com
• SAP BTP ABAP Environment Documentation: https://help.sap.com/docs/btp
• KIIT Kareer School — SAP ABAP Developer Course Material (IS30040), 2026
• SAP Standard Table Reference (VBAK, VBAP, KNA1) — SE11 Data Dictionary
• GitHub Repository: https://github.com/Being-Harshil/SAP-ABAP-Capstone-Project

NAME: HARSHIL MAINRA	**ROLL NO: 2306028**	**BRANCH: CSE (SAP ABAP)**